

Edge Geography is XNLP-hard for Pathwidth and in XP for Tree-Partition Width

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Edge Geography

One of the most famous PSPACE-complete problems.

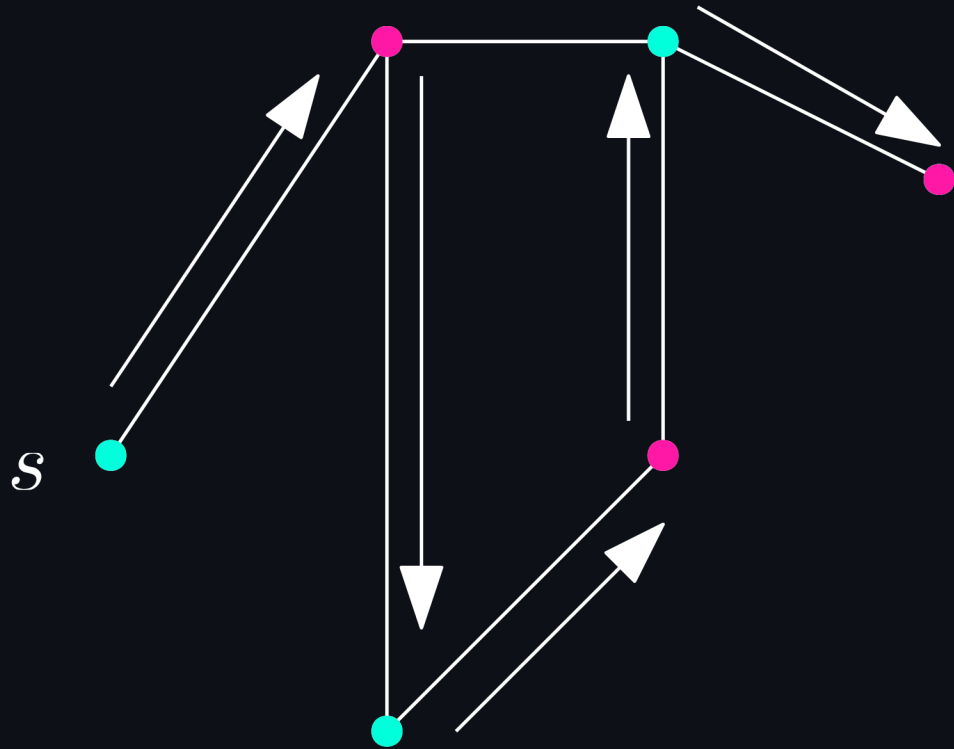
A token starts at a vertex s .

Players alternate:

1. choose an unused edge from the current vertex,
2. move the token along it,
3. delete that edge.

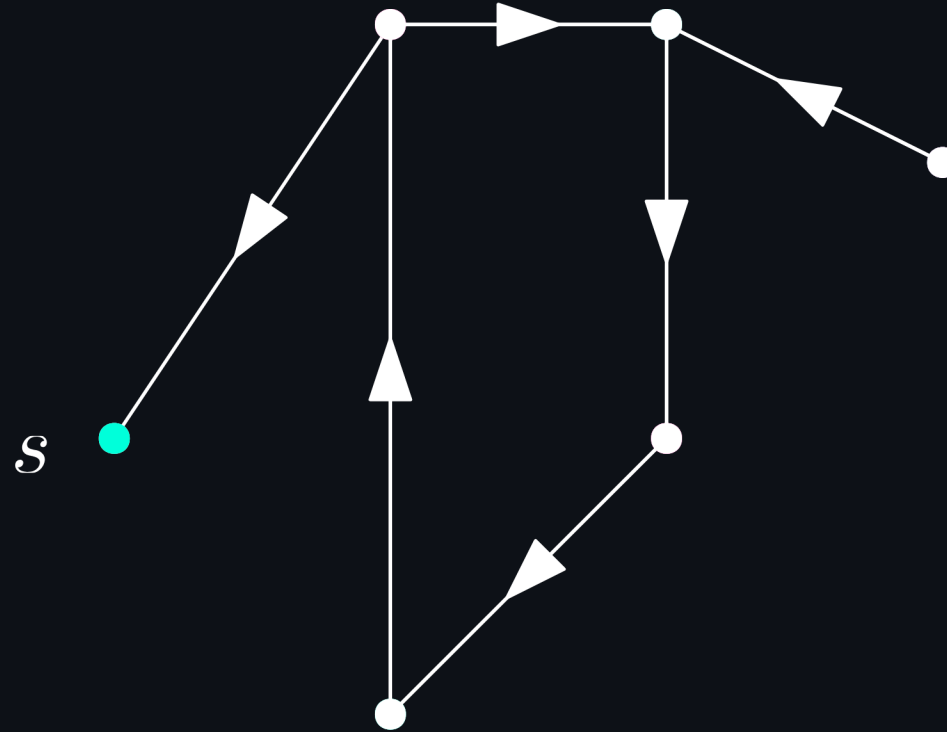
The first player unable to move loses.

Undirected Edge Geography



Player 1 wins!

Directed Edge Geography



Player 2 wins!

Parameterizing the problem

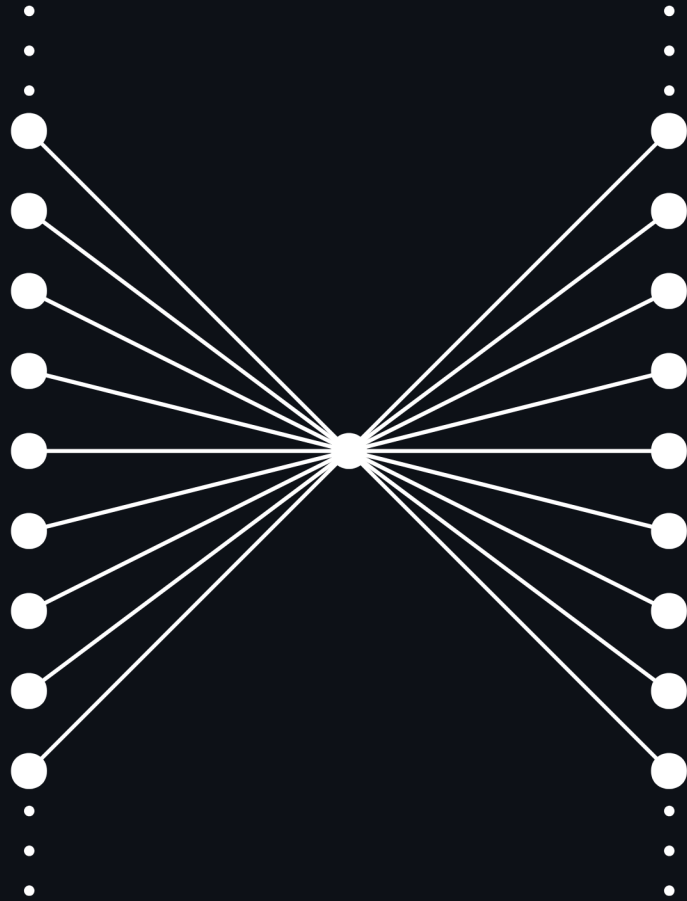
Theorem (Bodlaender, modern language)

(Undirected) Edge Geography is solvable in time $f(k, d) \cdot O(n)$ where k is the treewidth of the graph and d is the maximum degree of any vertex in the graph.

Exponential time becomes "linear time" when two parameters are controlled!

What if we allow d to be unbounded?

Arbitrarily many crossings



when $\Delta(G)$ is unbounded!

Our first result

Theorem

Directed Edge Geography is XNLP-hard parameterized by pathwidth.

XNLP "rules out" fixed-parameter tractability.

Bounded pathwidth $\Rightarrow$ bounded treewidth.

Pathwidth, and thus treewidth, is not enough!

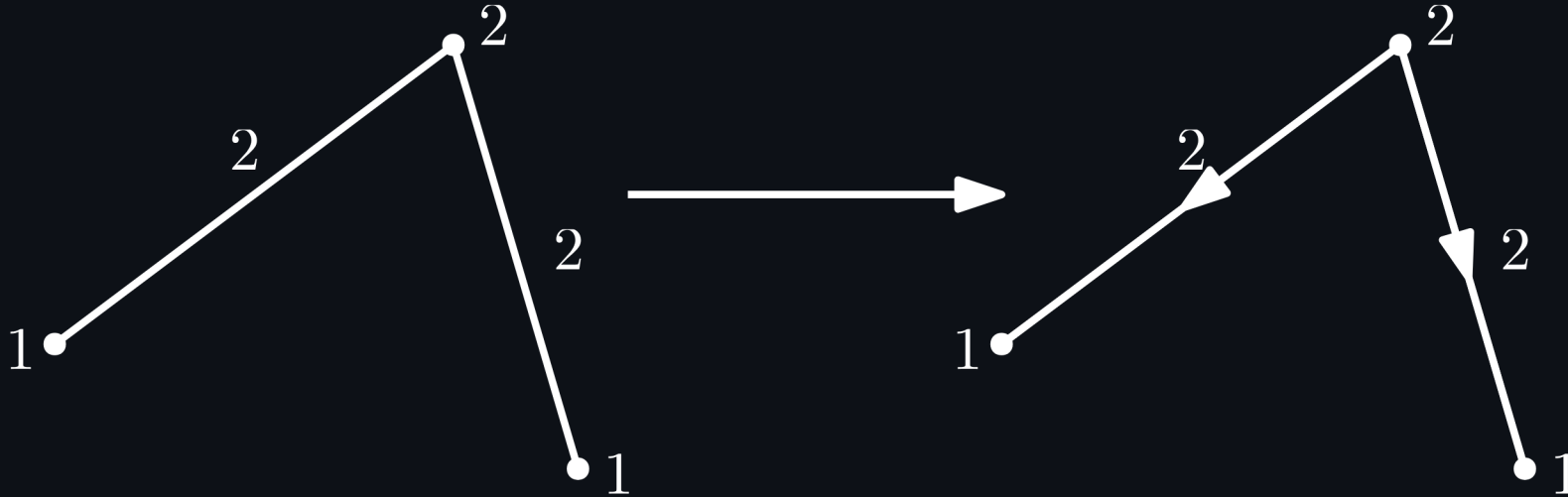
This result transfers to Undirected Edge Geography in a simple way!

Source problem: Chosen Maximum Outdegree

Does there exist an orientation of every edge so that each vertex respects its budget.

$$\sum_{e \in \theta_w(v)} w(e) \leq t(v) \quad \forall v$$

FIX THIS IMAGE IT IS WRONG



Known XNLP-complete when parameterized by pathwidth.

Reduction idea in two phases

Phase I — Player 1 encodes an orientation

For each input edge $e = \{x, y\}$, Player 1 chooses one of two sides.

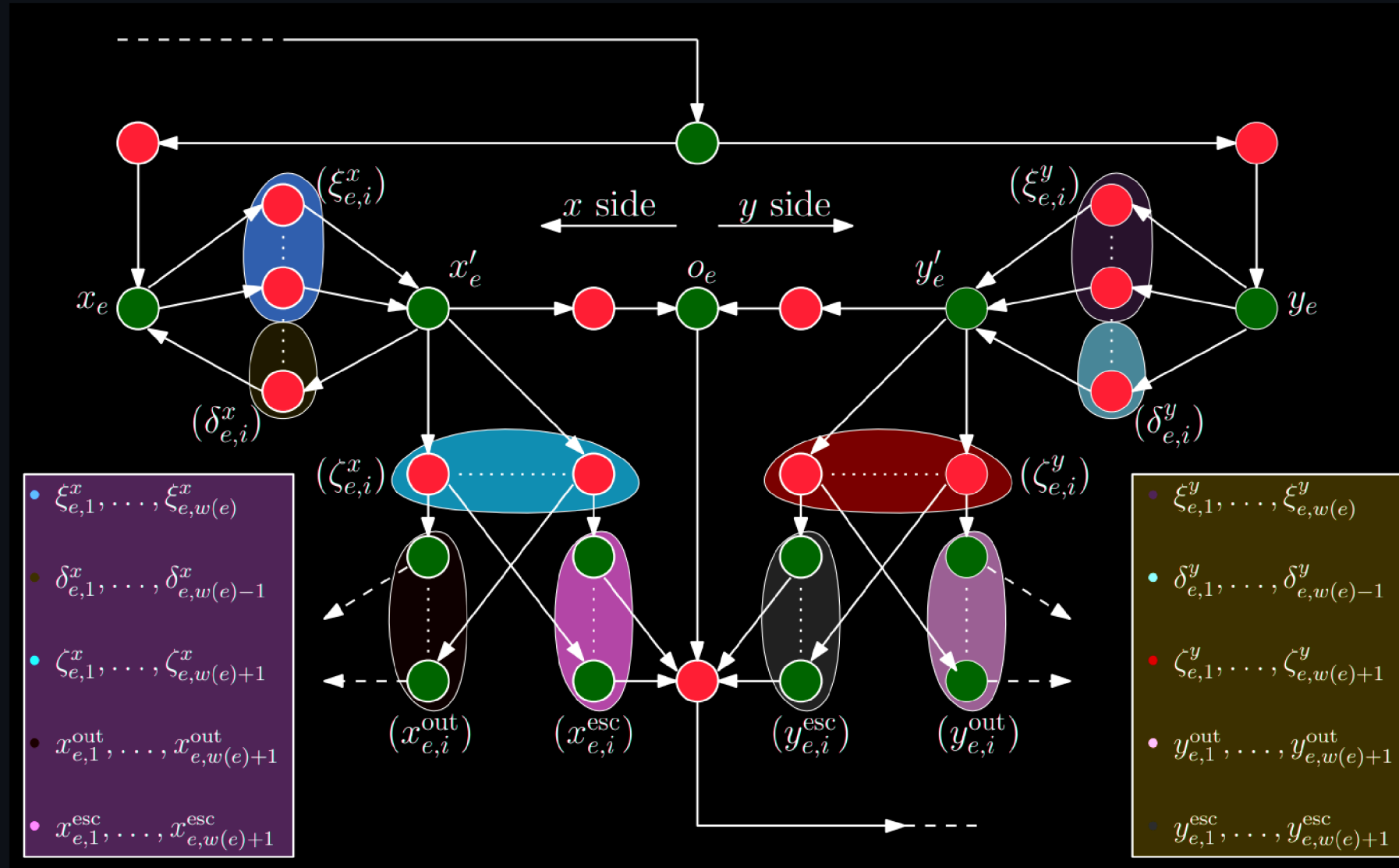
$$\text{choose the } y\text{-side} \iff x \rightarrow y$$

Phase II — Player 2 challenges a vertex

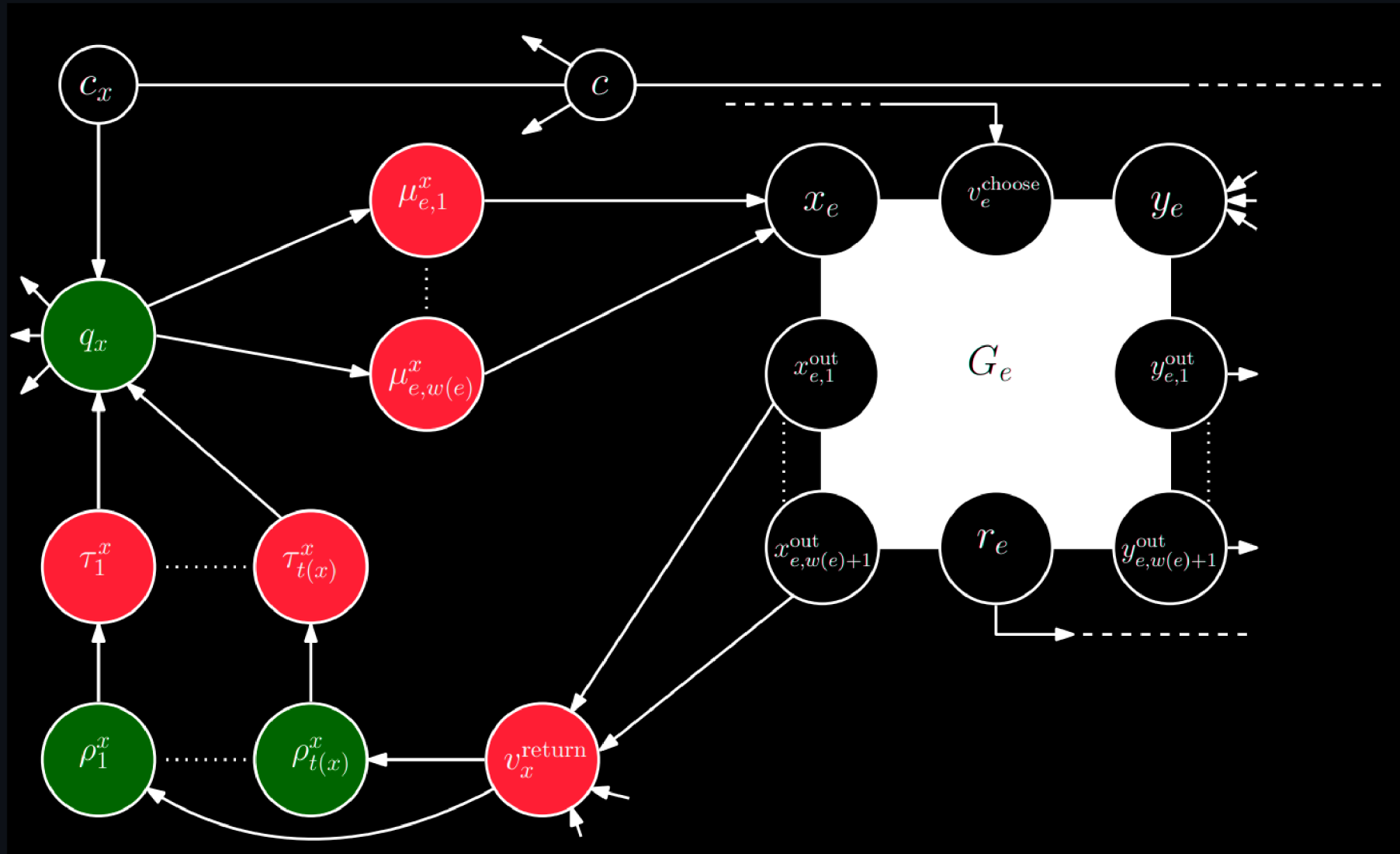
Player 2 chooses x and tests the total weight oriented out of x .

If outgoing weight $> t(x)$, Player 2 has one excursion too many.

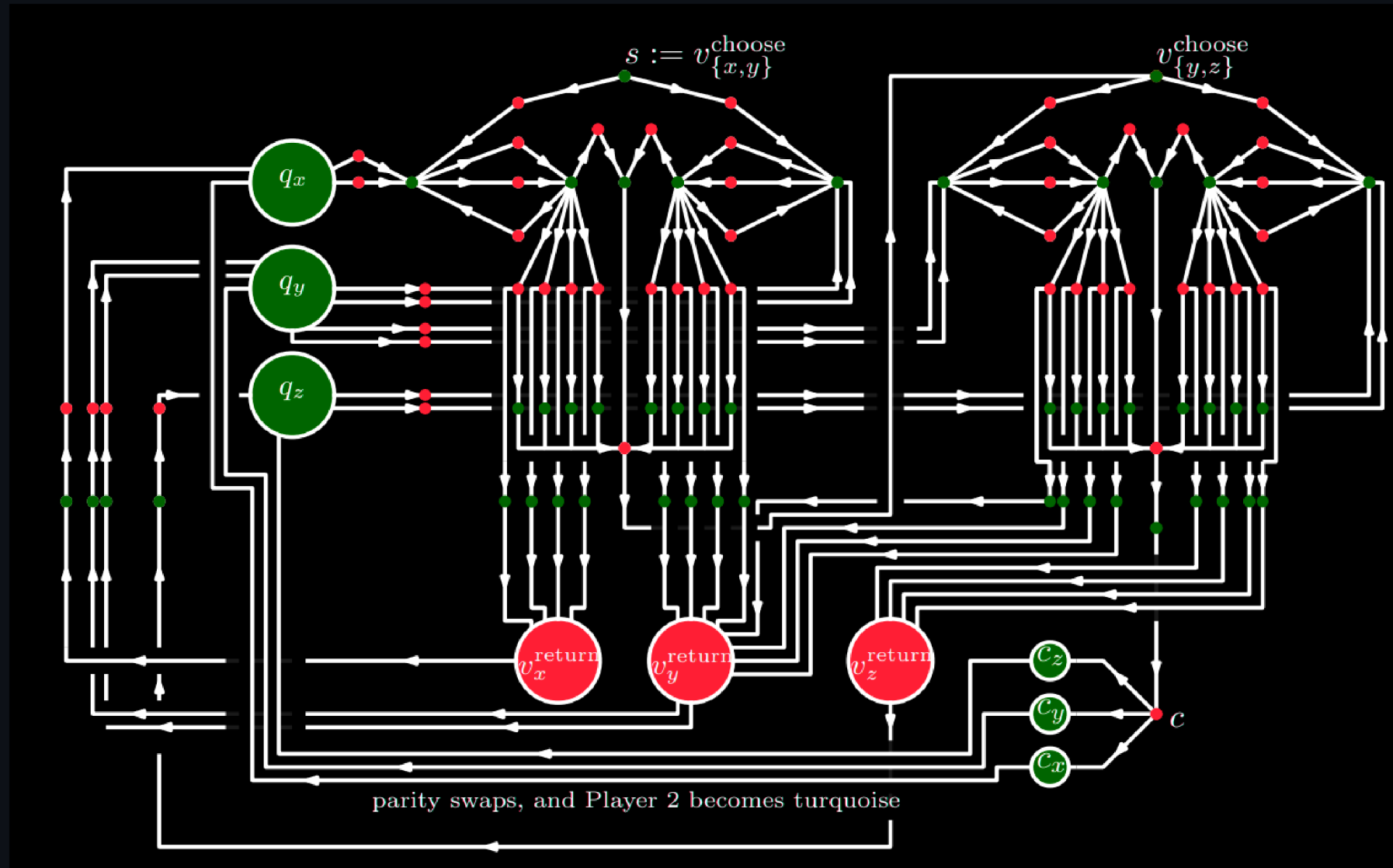
Phase 1



Phase 2



The full reduction for a small source instance



Why correctness follows

Feasible orientation

$$\sum_{e \in \theta_\omega(x)} w(e) \leq t(x) \text{ for every } x.$$

Player 2 eventually runs out of non-losing challenges.

Player 1 wins

Infeasible orientation

For some x ,

$$\sum_{e \in \theta_\omega(x)} w(e) > t(x).$$

Player 2 chooses this x and survives one challenge too many.

Player 2 wins

Why pathwidth is preserved

Process an input edge $e = \{x, y\}$ when x and y coexist in the source path decomposition.

Keep only the necessary anchors alive.

Introduce auxiliary gadget vertices locally, then forget them immediately.

$$\text{pw}(G') = O(k)$$

Directed to undirected

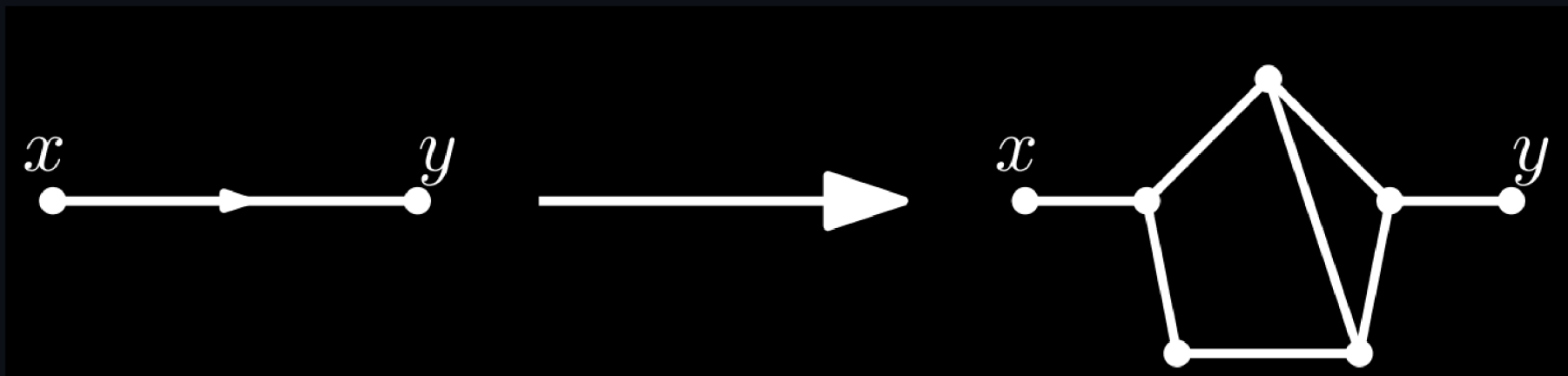
Each directed edge is replaced by a constant-size pseudoarc.

$u \rightarrow v \rightsquigarrow$ undirected pseudoarc

Safe traversal works from u to v .

Entering the wrong way is losing under optimal play.

Directed hardness transfers to Undirected Edge Geography.



XP membership for width of treepartition

A rooted tree partition groups vertices into bags.

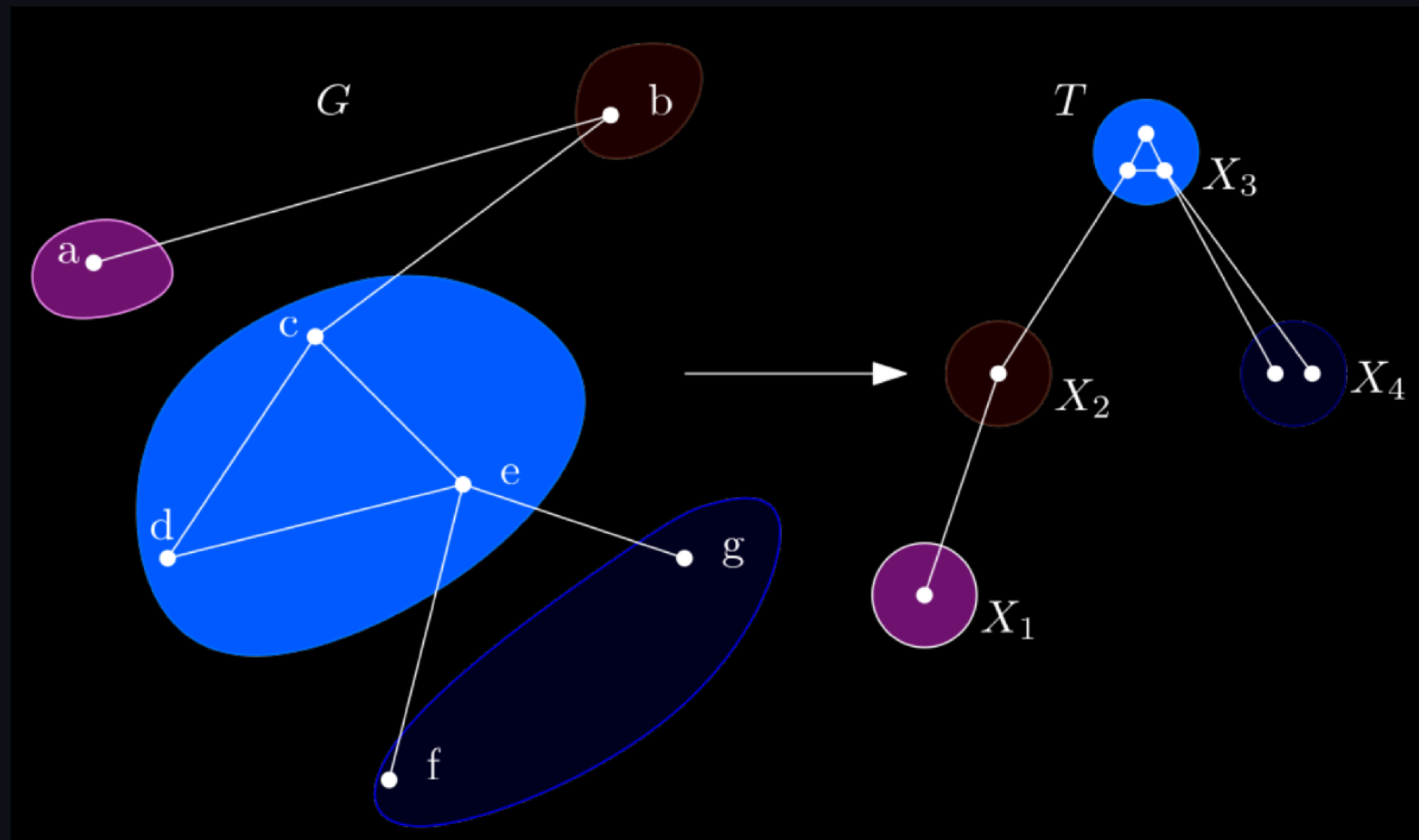
Edges are either:

- inside one bag, or
- between adjacent bags of the tree.

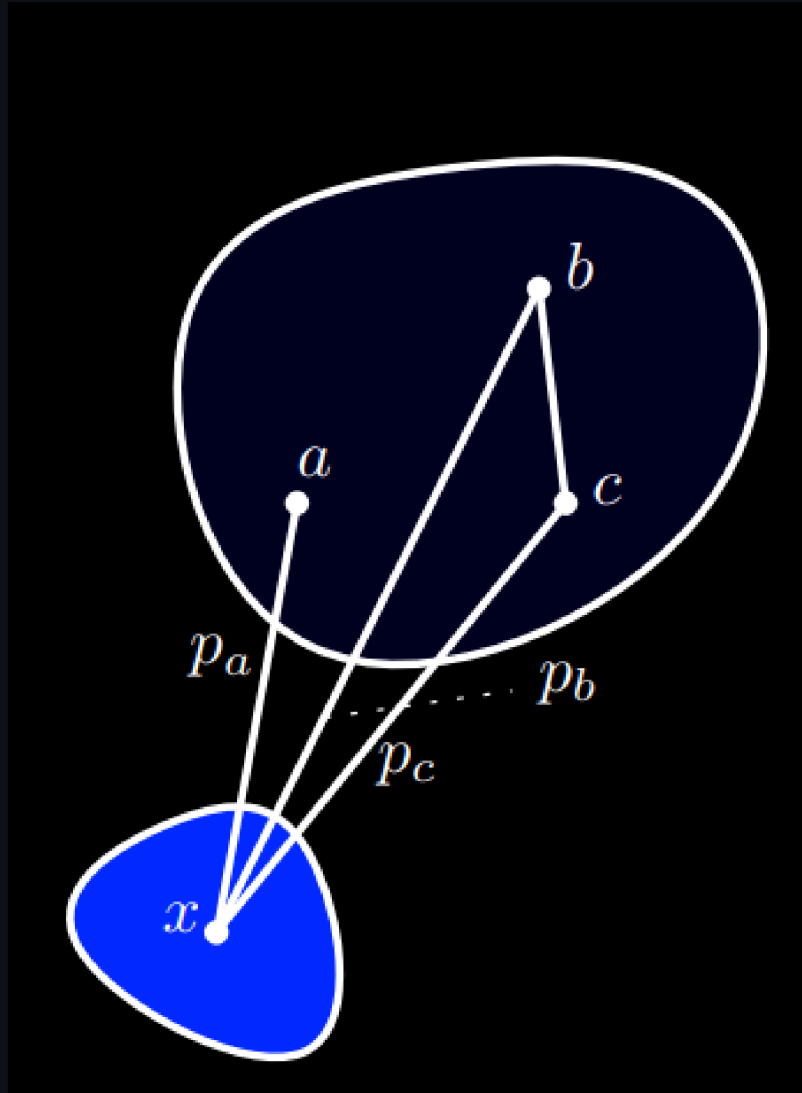
For width k , a parent-child cut has at most k^2 ports (on simple graphs).

Theorem

Undirected Edge Geography on simple graphs, given together with a rooted tree partition of width k , is solvable in XP time parameterized by k .



A solved child conveys a boolean function



The child exports a response

$$\Phi_a(\nu_b, \nu_c) = \nu_b \wedge \nu_c.$$

ν_b and ν_c are the values of the two possible continuations in the parent.

The cut has $O(k^2)$ ports, so the truth table of Φ_a has size only $f(k)$.

Returning through p_b also exports the smaller residual type σ_c .

Bottom-up symbolic minimax

leaf $\longrightarrow$ interface type $\longrightarrow$ parent $\longrightarrow \dots \longrightarrow$ root

1 · Leaves

For every entry port and local residual state, brute-force minimax **inside the bag**.

Treat possible returns to the parent as variables ν_p .

2 · Internal bags

Replace each solved child by its interface type σ .

Collect equal types into counters m_σ and run local minimax.

3 · Root

There is no unknown continuation above the root.

Evaluate the final compressed state to obtain the winner.

Entering a solved child is a **table lookup**; returning is a **counter update**.

Every move spends something forever

Once the children have been replaced by their types, resources can only decrease.

Inside the bag

One unused edge
disappears.

Child returns

Its entry and return ports
are gone.

Play exits upward

One parent port
disappears.

resources decrease $\implies$ no cycles $\implies$ reverse minimax

Implementation check.

The XP solver does indeed agree with separate brute-force minimax
on randomly generated instances.

Child counters comprise the worst factor

The n -dependent part

For fixed k , there are t_k possible child types.

Each counter lies in $\{0, \dots, n\}$, so

$$m \in \{0, \dots, n\}^{t_k} \implies \#m \leq (n+1)^{t_k} = n^{O(t_k)}.$$

The other factors

Unused ports and internal edges contribute

$$2^{k^2} \cdot 2^{\binom{k}{2}} = 2^{O(k^2)}.$$

All other boundary data depends only on k ; collect it into $g(k)$.

$$\underbrace{n}_{\text{bags}} \cdot \underbrace{g(k)}_{\text{includes } 2^{O(k^2)}} \cdot \underbrace{n^{O(t_k)}}_{\text{counter states}} = n^{f(k)}.$$

Laughably bad in practice

Let T_r be the number of ambient types on an interface with r ports.

ports r	0	1	2	3	4
ambient types T_r	1	2	676	$\approx 2^{768}$	$\geq 2^{2^{4058}}$

Illustrative ambient count obtained by unwinding Definition 35.

The number of realizable types may be vastly smaller.

A width- k cut may have k^2 ports.

The crude bound is a tower of exponentials of height $O(k^2)$.

Takeaway

Pathwidth

small separators

**but residual history is
uncontrolled**

XNLP-hard

Tree-partition width

bounded interfaces

residual play can be compressed

XP

Open: Is Edge Geography in XP parameterized by pathwidth alone?